Facultad de Defensa Nacional

Carrera Lic. En Ciberdefensa

Análisis Matemático I

Profesor a cargo: Lic. Nelson Monzón

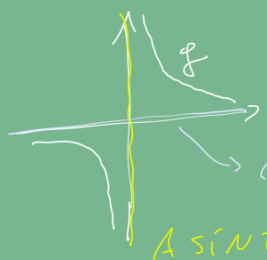
Profesora adjunta: Prof. Maela Lombardi

Año: 2025

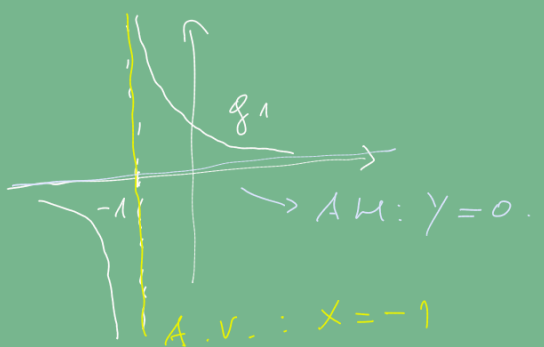
Carga horaria: 64

Ej: $f(x) = -2 - \frac{1}{x+1}$

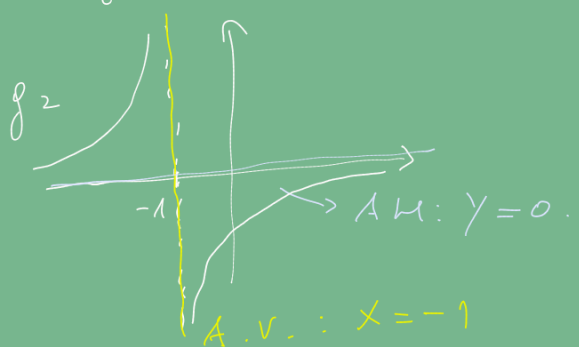
$g(x) = \frac{1}{x}$



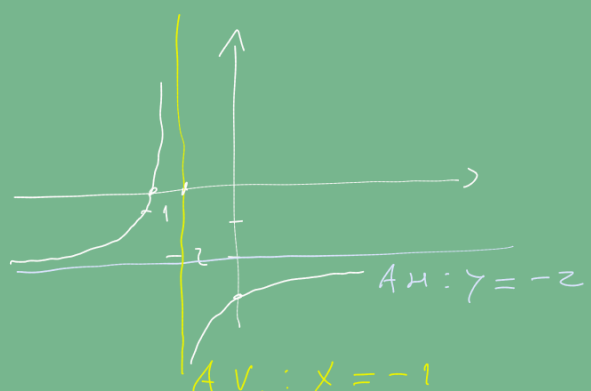
, $g_1(x) = \frac{1}{x+1} = g(x+1)$



$g_2(x) = \frac{-1}{x+1} = -g_1(x)$



$f(x) = -2 - \frac{1}{x+1} = -2 + g_2(x)$



OBS:

$f(x-a)$ → $(a>0)$ →, $(a<0)$ ←

$f(x)+a$ → $(a>0)$ ↑, $(a<0)$ ↓

$a \cdot f(x)$ → $a>1$ ↕, $0<a<1$ ↓↑

$-f(x)$ ⟶ ↑↓

$f(-x)$:



$f(a.x)$:

$\longleftrightarrow$ $(0 < a < 1)$

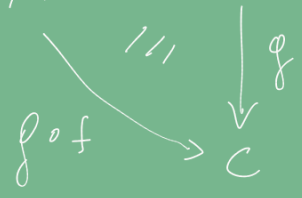
$\longrightarrow\ \longleftarrow$ $(a > 1)$

$f(-5x)$

COMPOSICIÓN DE FUNCIONES:

$f : A \longrightarrow B$, $g : B \longrightarrow C$

$g \circ f(x) = g(f(x))$



EJ: $g : \mathbb{R} \longrightarrow \mathbb{R}$, $g(x) = x^2$

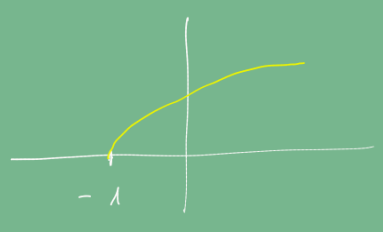
$f : \mathbb{R} \longrightarrow \mathbb{R}$, $f(x) = x + 1$

$g \circ f(x) = g(f(x)) = g(x+1) = (x+1)^2$

$f \circ g(x) = f(g(x)) = f(x^2) = x^2 + 1$

OBS: EN GENERAL, $g \circ f \neq f \circ g$

EJ: $f(x) = \sqrt{x+1}$



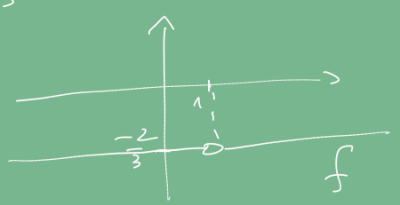
FUNCIÓN HOMOGRÁFICA:

$$f(x) = \frac{ax+b}{cx+d}, \quad c \neq 0, \quad ad \neq bc$$

Ej: $f(x) = \dfrac{2x-2}{-3x+3}$ $\quad 2 \cdot 3 = -2 \cdot (-3)$

NO CUMPLE $ad = bc$.

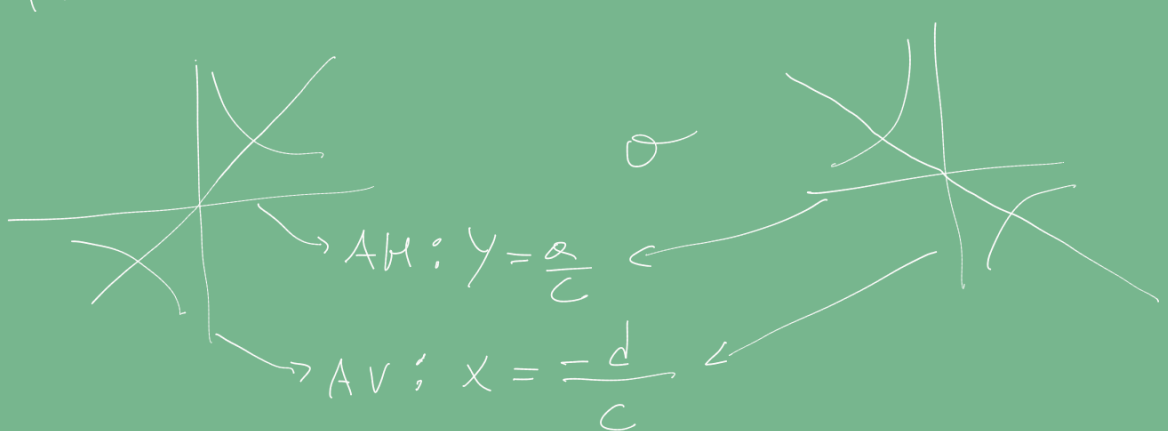
$$f(x) = \frac{2}{-3}\,\frac{x-1}{x-1} = -\frac{2}{3} \quad \text{si } x \neq 1$$



FUNCIÓN HOMOGRÁFICA:

$$f(x) = \frac{ax+b}{cx+d}, \quad c \neq 0, \quad ad \neq bc$$

TIPOS DE GRÁFICO:



Ej: $f(x) = \dfrac{2x-1}{1 \cdot x+3}$

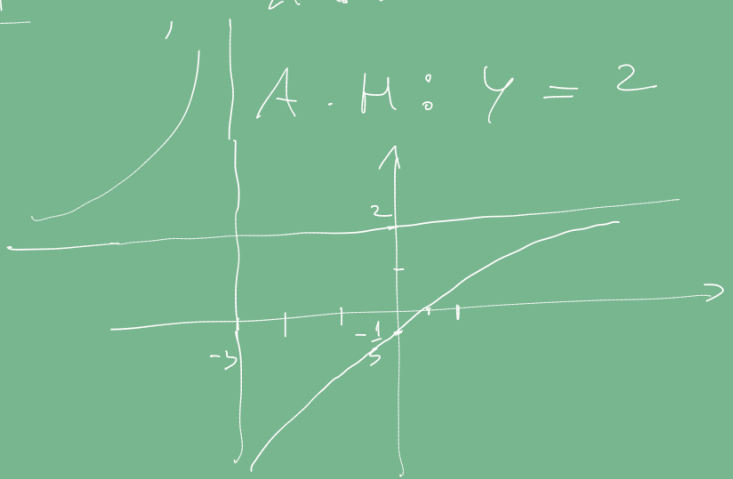
$c = 1 \neq 0$ ✓

$2 \cdot 3 \neq (-1) \cdot 1$ ✓

$f(0) = \dfrac{-1}{3}$

A.V.: $x = -3$

A.H: $y = 2$



$$f(x) = \frac{2x-1}{x+3} = \frac{2(x+3)-7}{x+3} = 2 - \frac{7}{x+7}$$

$$\begin{array}{r|l} \cancel{2x} - 1 & x+3 \\ \hline \cancel{2x}+6 & 2 \\ \hline -7 & \end{array} \qquad \Rightarrow 2x-1 = 2(x+3)-7$$

$\frac{1}{x}$ ↑ ↓

$$f(x) = A + \frac{B}{x-D} \qquad (B \neq 0)$$

A.M.: $y = A$

A.V: $x = D$